UTMSTACK OBSERVATORY

User Manual

Version 1.16 · October 2026

UTMStack Observatory — User Manual

Version 1.16 · October 2026

UTMStack Observatory is a read-only operations dashboard for a UTMStack v11 SIEM. It sits next to UTMStack, reads its OpenSearch data, and turns it into one screen a security team, an IT manager or a NOC wall can understand at a glance: which data sources are reporting, how many events and alerts are coming in, where firewall traffic and attacks come from on a live world map, and whether the SIEM itself is healthy.

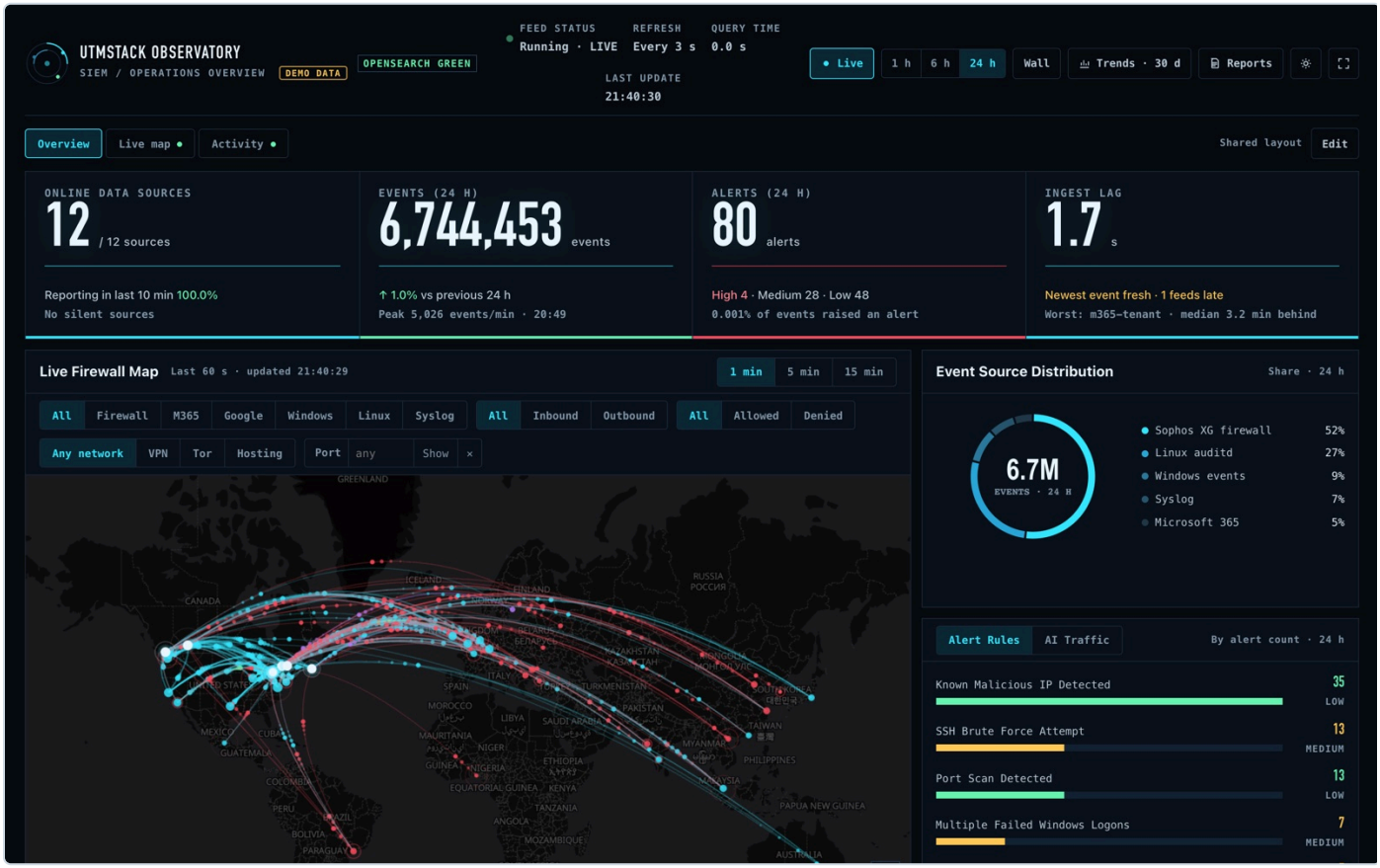
It never changes anything in UTMStack. It only reads.

> Every screenshot and clip in this manual was taken in **demo mode**, which uses a fictional organisation and addresses from the reserved documentation ranges. No real customer data is shown. See Demo mode.

Contents

1. At a glance
2. Signing in
3. The header
4. Overview screen
5. Live map
6. Alerts
7. Data sources and ingest lag
8. SIEM health
9. Trends and history
10. Reports
11. Screens, widgets and Wall mode
12. Settings panel
13. Demo mode
14. Installation and upgrades
15. Configuration reference
16. Security and privacy
17. Troubleshooting
18. Third-party data and licences

1. At a glance



You want to know…	Look at
Is every log source still sending?	**Online data sources** card
How busy is the SIEM, and is that normal?	**Events** card (with change against the previous period)
Is anything being flagged?	**Alerts** card and the **Alert Rules** panel
Are logs arriving late?	**Ingest lag** card
Where is traffic going, and who is attacking us?	**Live map**
Is UTMStack's database healthy?	The **OpenSearch** status badge next to the title
How does this week compare with last week?	**Trends**
Something to send to management	**Reports** (PDF or HTML)

The dashboard refreshes itself every few seconds. **Live** in the header means you are looking at current data.

2. Signing in

Open the address your administrator gave you, for example `http://siem-server:8765` (or `https://…` if TLS is on). Your browser asks for a user name and password. The default user is `admin`; the installer generates the password and prints it once at the end of the install.

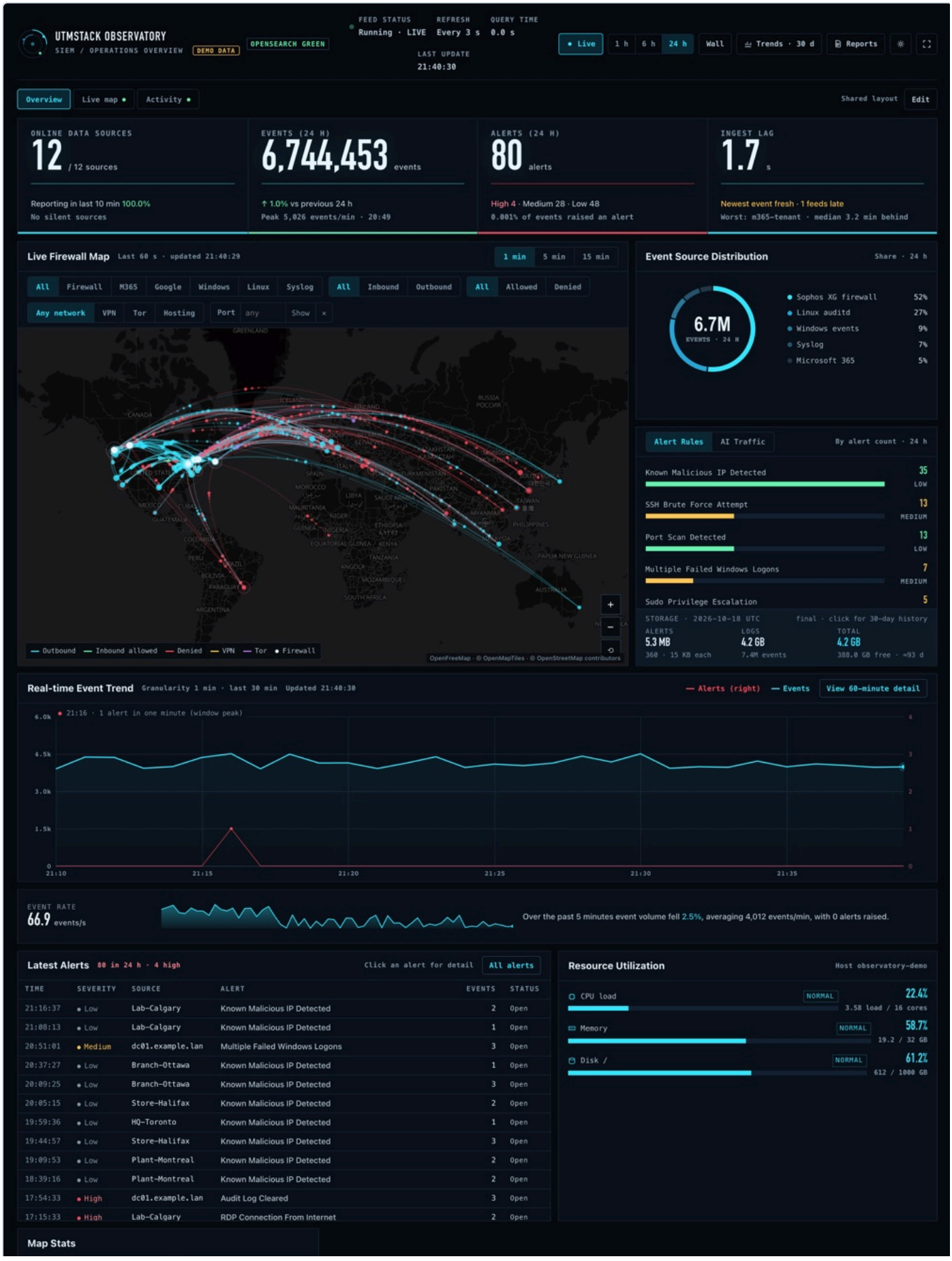
- After **5 wrong passwords** from the same address, that address is locked out for **15 minutes**, even if the next attempt is correct. Both numbers can be changed (see Configuration reference).
- To change the password, use the Settings panel or `sudo ./install.sh password` on the server.
- To sign out, close the browser. Basic authentication has no sign-out button.

3. The header

Control	What it does
Logo	Shows your support contact (name, email, phone, website) and the installed version.
OpenSearch status badge (green / yellow / red)	Health of UTMStack's database. Click it to see what is causing a yellow or red state and how to fix it. See SIEM health.
Feed status / Last update	Whether the dashboard is receiving data and when it last refreshed.
Live · 1 h · 6 h · 24 h	Time range for the counters, the event-source donut, the top alert rules and the alert list. **Live** keeps the default 24-hour view moving.
Wall	Full-screen NOC mode with site names hidden. See Wall mode.
Trends · 30 d	30 days of history with daily and weekly comparisons.
Reports	Generated PDF/HTML reports and a **Generate now** button.
Gear	Settings panel.
Full screen	Uses the whole monitor without entering Wall mode.

Below the header are the **screen tabs** (Overview, Live map, Activity, and any screens you add) and the **Edit** button for changing the layout.

4. Overview screen


UTMSTACK OBSERVATORY
SIEM / OPERATIONS OVERVIEW
DEMO DATA
OPENSEARCH GREEN
FEED STATUS
Running · LIVE
REFRESH
Every 3 s
QUERY TIME
0.0 s
LAST UPDATE
21:40:30
Live
1 h
6 h
24 h
Wall
Trends · 30 d
Reports
Overview
Live map
Activity
Shared layout
Edit
ONLINE DATA SOURCES
12 / 12 sources
Reporting in last 10 min 100.0%
No silent sources
EVENTS (24 H)
6,744,453 events
↑ 1.0% vs previous 24 h
Peak 5,026 events/min · 20:49
ALERTS (24 H)
80 alerts
High 4 · Medium 28 · Low 48
0.001% of events raised an alert
INGEST LAG
1.7 s
Newest event fresh · 1 feeds late
Worst: m365-tenant · median 3.2 min behind
Live Firewall Map
Last 60 s · updated 21:40:29
1 min
5 min
15 min
All
Firewall
M365
Google
Windows
Linux
Syslog
All
Inbound
Outbound
All
Allowed
Denied
Any network
VPN
Tor
Hosting
Port
any
Show
Outbound
Inbound allowed
Denied
VPN
Tor
Firewall
OpenFreeMap · © OpenMapTiles · © OpenStreetMap contributors
Event Source Distribution
Share · 24 h
6.7M
EVENTS · 24 H
Sophos XG firewall 52%
Linux auditd 27%
Windows events 9%
Syslog 7%
Microsoft 365 5%
Alert Rules
AI Traffic
By alert count · 24 h
Known Malicious IP Detected 35 LOW
SSH Brute Force Attempt 13 MEDIUM
Port Scan Detected 13 LOW
Multiple Failed Windows Logons 7 MEDIUM
Sudo Privilege Escalation 5
STORAGE · 2026-10-18 UTC
final · click for 30-day history
ALERTS 5.3 MB 360 · 15 KB each
LOGS 4.2 GB 7.4M events
TOTAL 4.2 GB 388.0 GB free · ≈93 d
Real-time Event Trend
Granularity 1 min · last 30 min Updated 21:40:30
Alerts (right)
Events
View 60-minute detail
21:16 · 1 alert in one minute (window peak)
6.0k
4.5k
3.0k
1.5k
0
21:10
21:15
21:20
21:25
21:30
21:35
EVENT RATE
66.9 events/s
Over the past 5 minutes event volume fell 2.5%, averaging 4,012 events/min, with 0 alerts raised.
Latest Alerts
80 in 24 h · 4 high
Click an alert for detail
All alerts
TIME SEVERITY SOURCE ALERT EVENTS STATUS
21:16:37 Low Lab-Calgary Known Malicious IP Detected 2 Open
21:08:13 Low Lab-Calgary Known Malicious IP Detected 1 Open
20:51:01 Medium dc01.example.lan Multiple Failed Windows Logons 3 Open
20:37:27 Low Branch-Ottawa Known Malicious IP Detected 1 Open
20:09:25 Low Branch-Ottawa Known Malicious IP Detected 3 Open
20:05:15 Low Store-Halifax Known Malicious IP Detected 2 Open
19:59:36 Low HQ-Toronto Known Malicious IP Detected 1 Open
19:44:57 Low Store-Halifax Known Malicious IP Detected 3 Open
19:09:53 Low Plant-Montreal Known Malicious IP Detected 2 Open
18:39:16 Low Plant-Montreal Known Malicious IP Detected 2 Open
17:54:33 High dc01.example.lan Audit Log Cleared 3 Open
17:15:33 High Lab-Calgary RDP Connection From Internet 2 Open
Resource Utilization
Host observatory-demo
CPU load NORMAL 22.4% 3.58 load / 16 cores
Memory NORMAL 58.7% 19.2 / 32 GB
Disk / NORMAL 61.2% 612 / 1000 GB
Map Stats

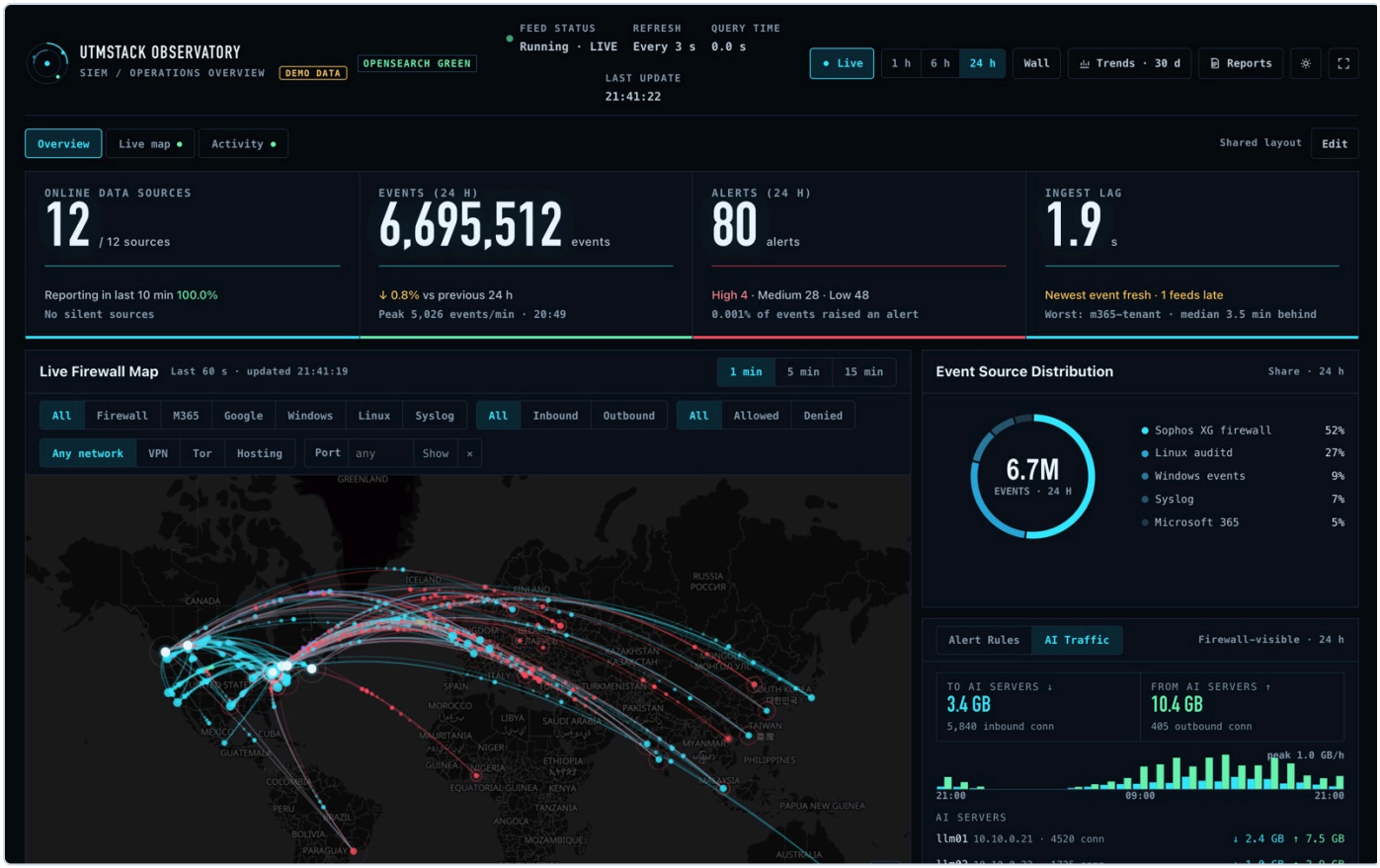


KPI cards (top row):

- **Online data sources**: sources that reported in the last 10 minutes, out of all known sources. *No silent sources* means everything is reporting. Click the card for the full source list.
- **Events**: events stored in the selected period, the change against the previous period, and the peak rate.
- **Alerts**: alerts raised in the period, split by severity (High / Medium / Low), and the share of events that raised an alert. Click it for the alert list.
- **Ingest lag**: how old the newest stored event is. A green value of a few seconds is normal. Click it for the per-source analysis.

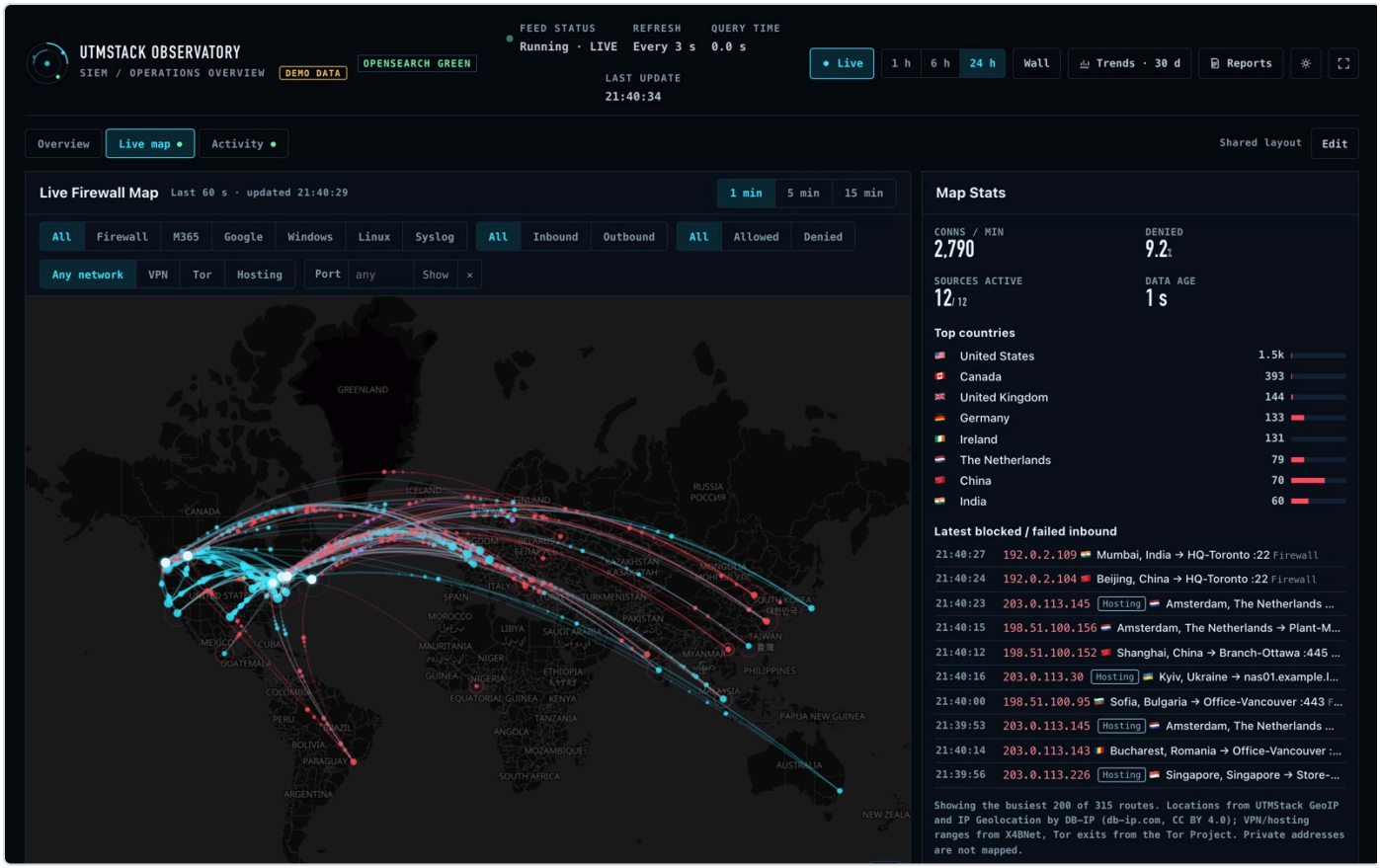
Panels (all movable, see Screens and widgets):

- **Live firewall map**: a compact version of the Live map.
- **Event source distribution**: a donut of events by source type (firewall, Linux, Windows, syslog, Microsoft 365…).
- **Alert Rules**: the rules raising the most alerts. **Click a rule** to open its alerts.
- **AI Traffic** (optional): bytes in and out of chosen servers, for example AI or GPU hosts, measured from firewall logs. Turned on by listing hosts in the Settings panel.
- **Alert timeline**, **storage**, **resources** and other widgets can be added from the widget catalogue.



5. Live map

The Live map draws every connection the firewalls log as an arc between the firewall and the remote city, and plots attackers seen in server and cloud logs.



Reading the map

Colour	Meaning
Cyan	Outbound: a local device connected out
Blue-green	Inbound and allowed
Red	Denied or failed (blocked by the firewall, or a failed login)
Yellow	From a known VPN range
Purple	From a Tor exit node
White dots	Your firewalls

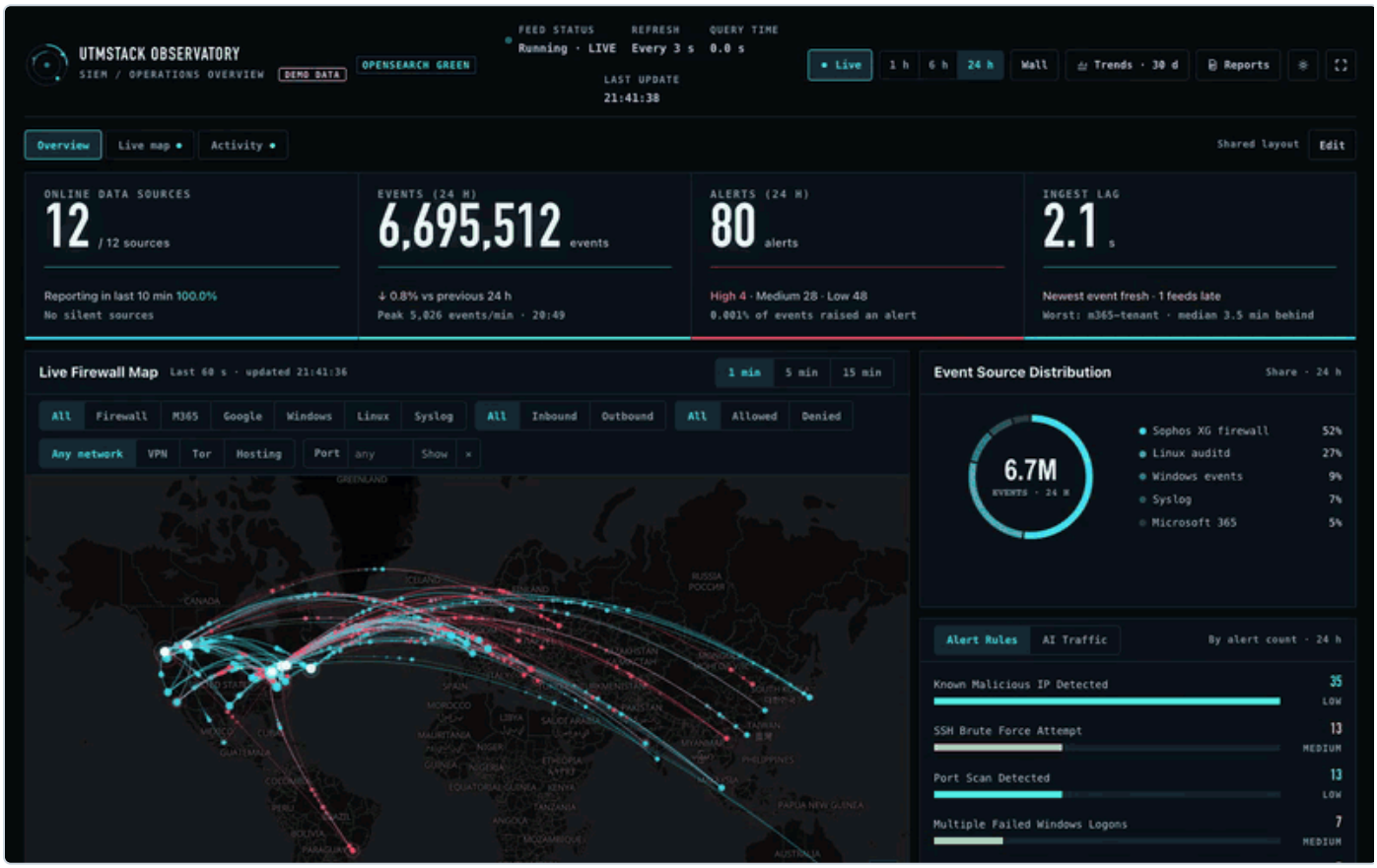
Arcs move in the direction of the traffic. The side panel shows connections per minute, the percentage denied, top countries, and **Latest blocked / failed inbound**, a ticker of the newest blocked connections with their source address, location, target and port. Badges on a row mark **VPN**, **Tor** or **Hosting** (datacenter) addresses. Hover a ticker row to highlight that location on the map.

Window and filters

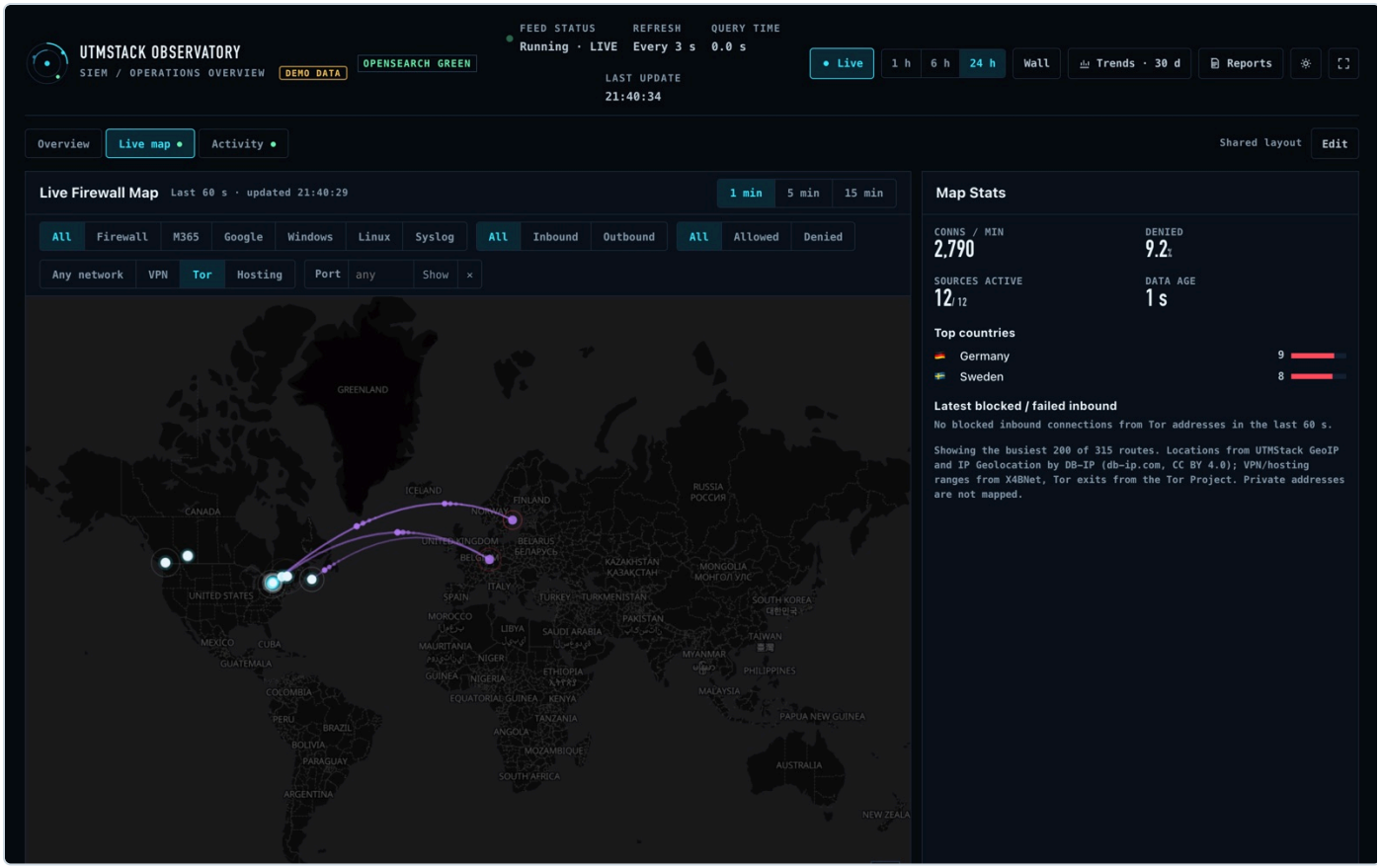
- **1 min · 5 min · 15 min**: how much recent traffic is drawn.
- **Source**: All, Firewall, M365, Google, Windows, Linux, Syslog. Only sources that carry IP addresses can be placed; a short bar at the top of the map says so when the selected source has nothing to show.
- **Direction**: All, Inbound, Outbound.

- **Outcome**: All, Allowed, Denied.
- **Network**: Any network, VPN, Tor, Hosting.
- **Port**: type a port (for example `22`, `3389`, `443`) and press **Enter** or **Show** to see only traffic to that port. ✕ clears it. Windows and Microsoft 365 events carry no port, so they drop out while a port is set.

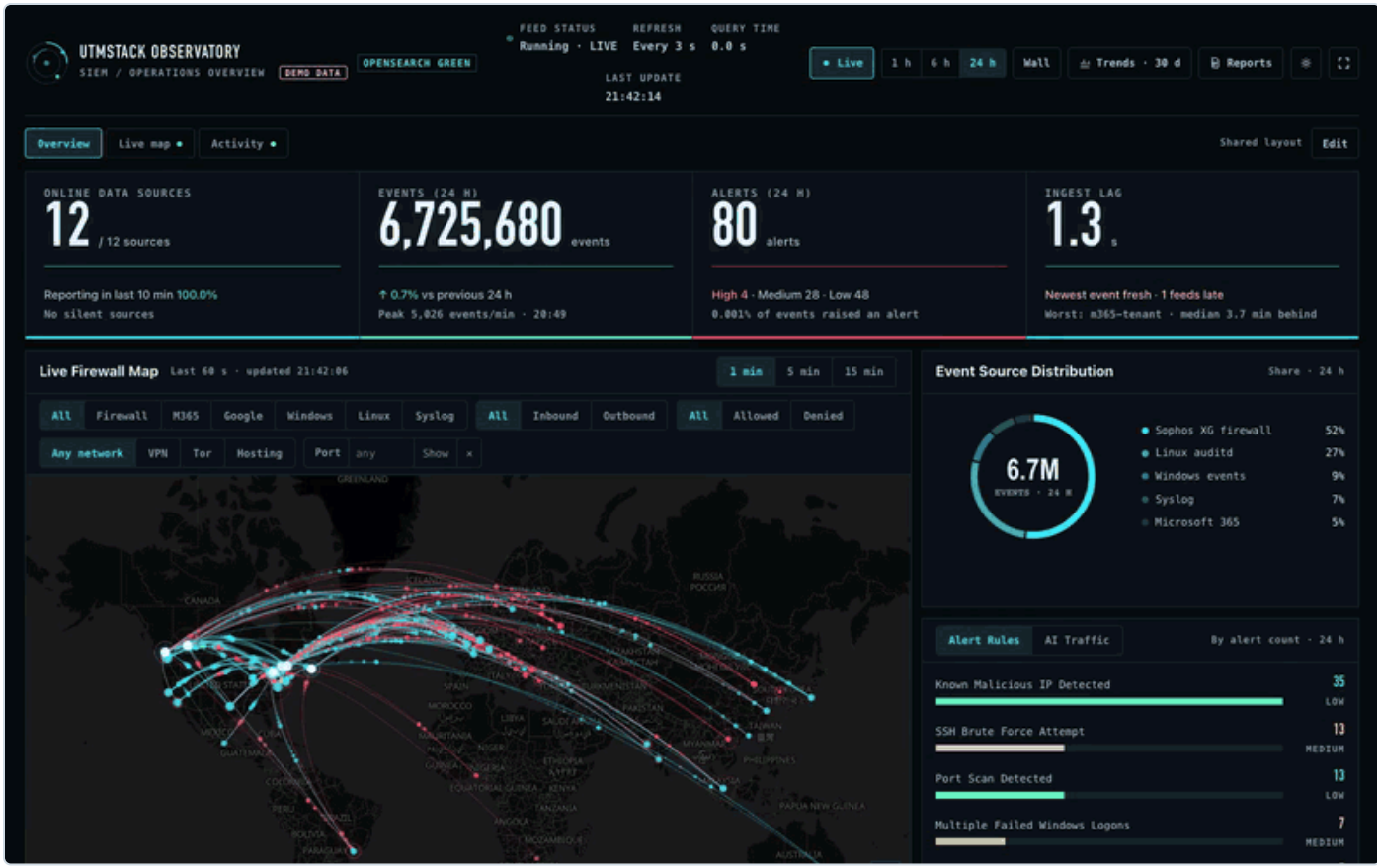
All filters combine. The side panel and ticker follow them.



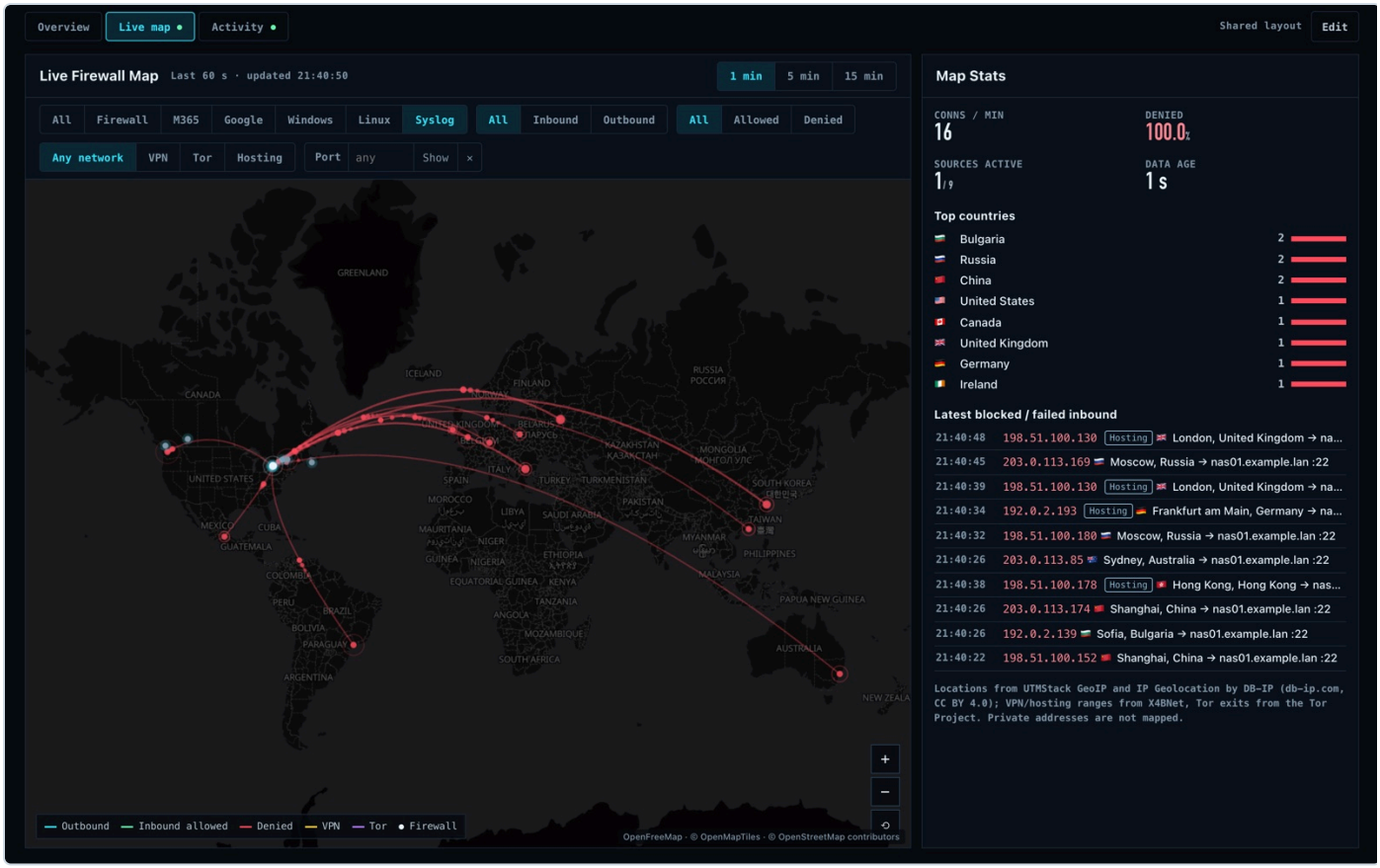
Tor traffic only:



Port search: where SSH (22) and RDP (3389) attempts come from:

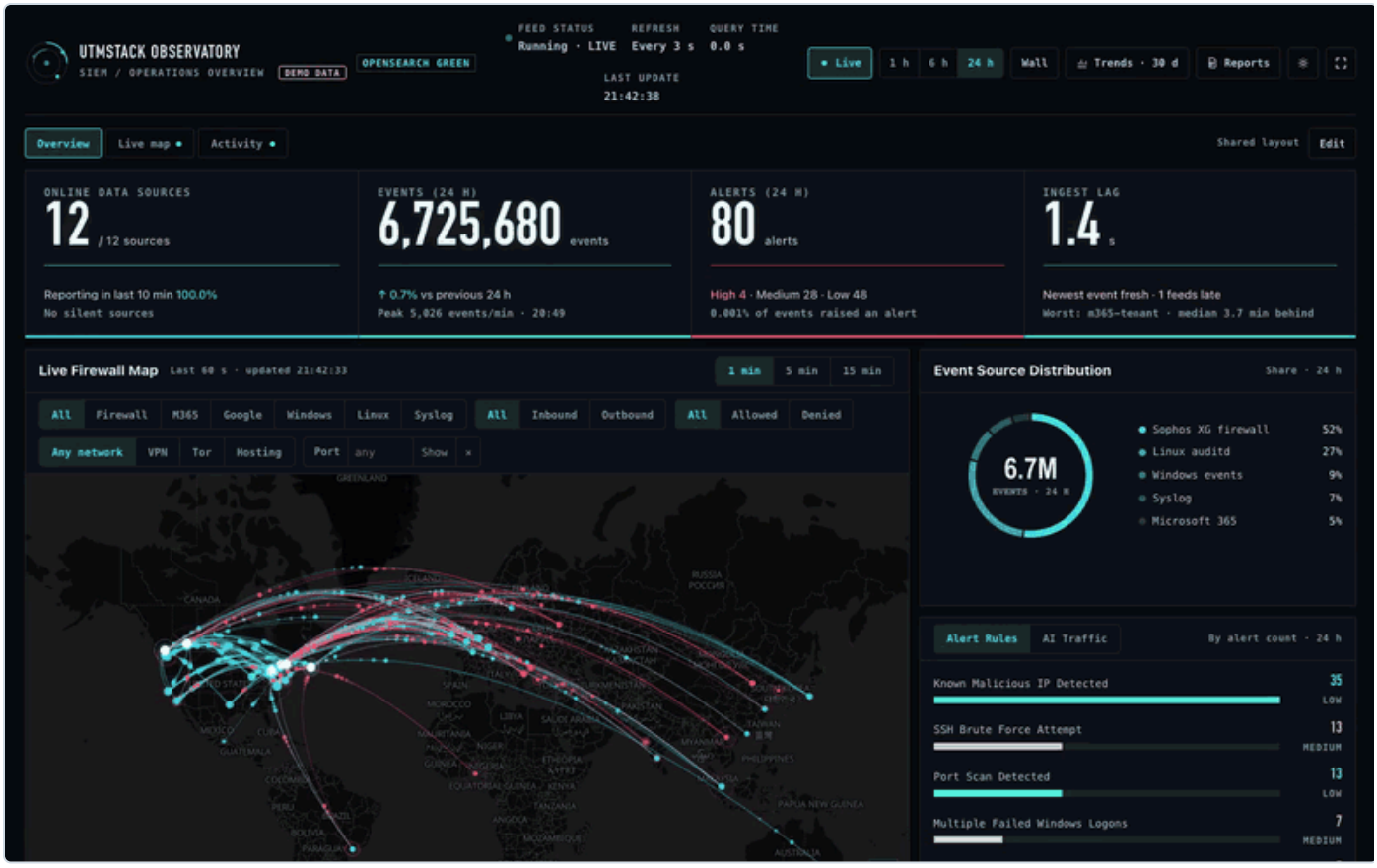


Attackers found in server logs (syslog / SSH):

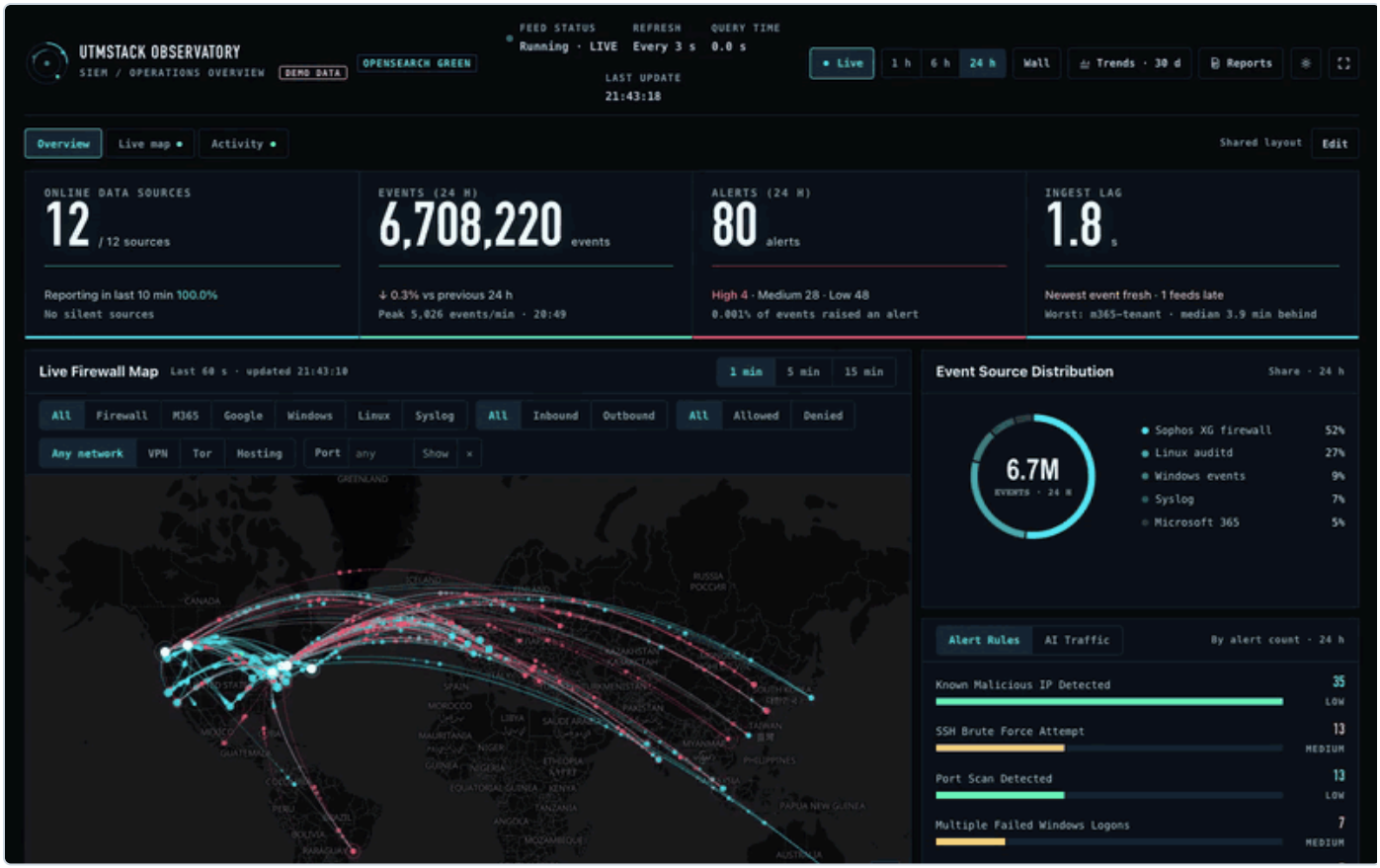


Zoom and street map

Use **+ / – / ⟲**, the mouse wheel, double-click or drag; on a phone, pinch and drag (a single tap shows a tooltip). Zoom in far enough and a street-level map appears underneath the arcs.



> IP-based location is city-level at best. The street map adds context; it does not pinpoint an attacker's building.



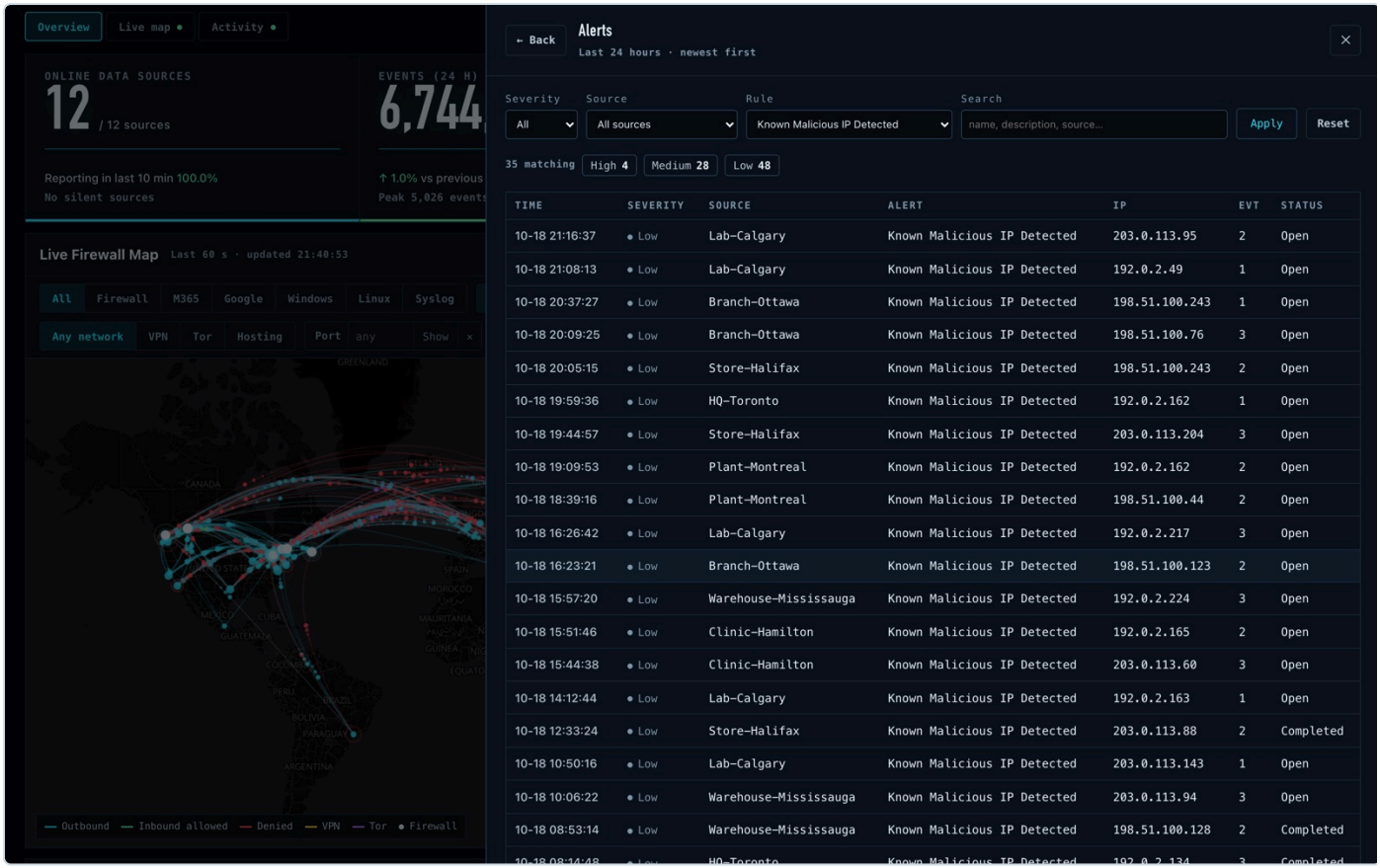
Where locations come from

1. UTMStack's own GeoIP enrichment, when present in the log.
2. A local copy of the DB-IP City Lite database, plus VPN, datacenter and Tor exit lists, downloaded and refreshed automatically by the dashboard.
3. Optionally, an online lookup (proxycheck.io) for addresses the local data cannot place, capped at a daily number (default 100, `0` turns it off).

Private addresses (10.x, 192.168.x, …) are never placed on the map and never sent anywhere.

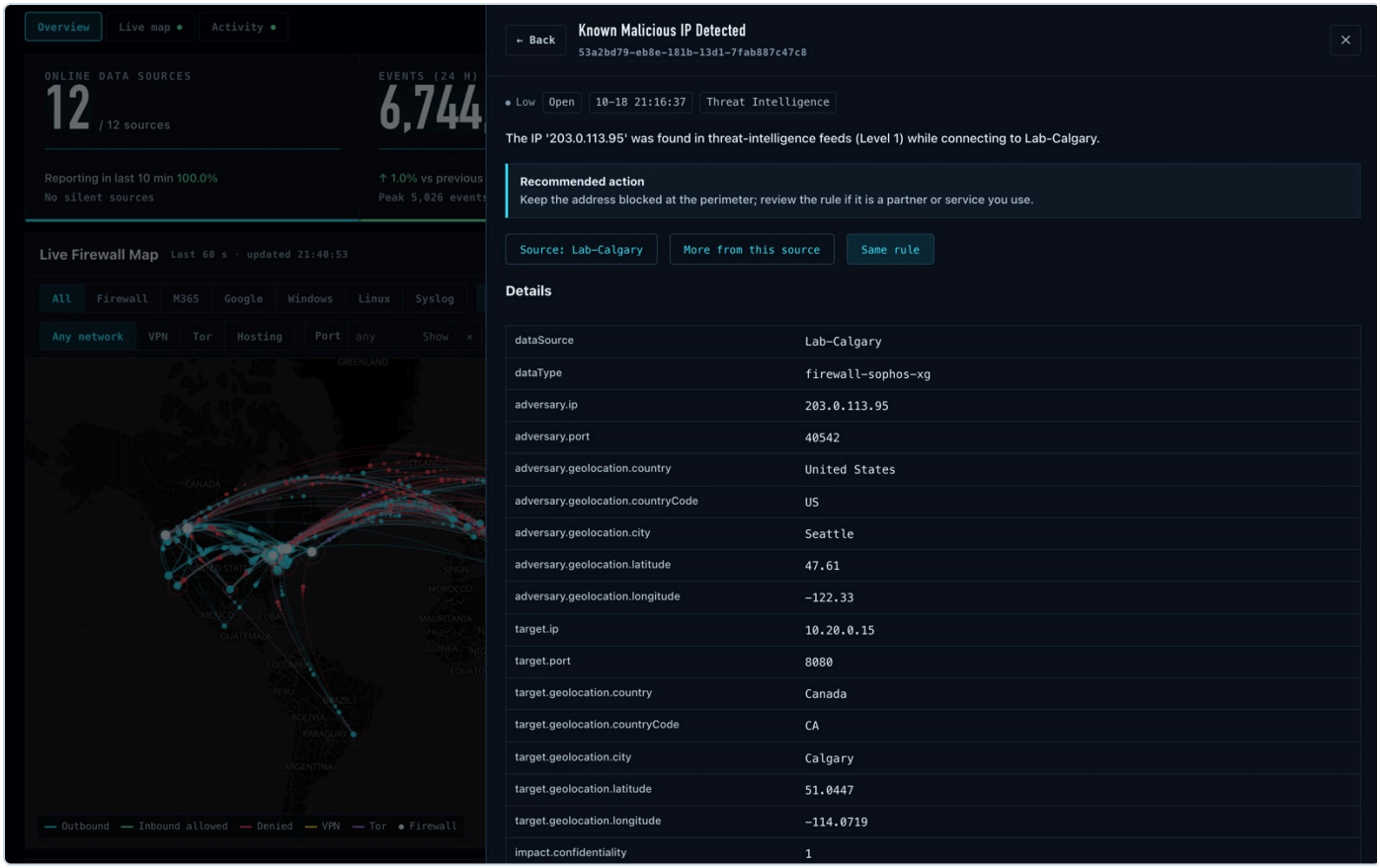
6. Alerts

Click a rule in **Alert Rules**, or the **Alerts** card, to open the alert list.

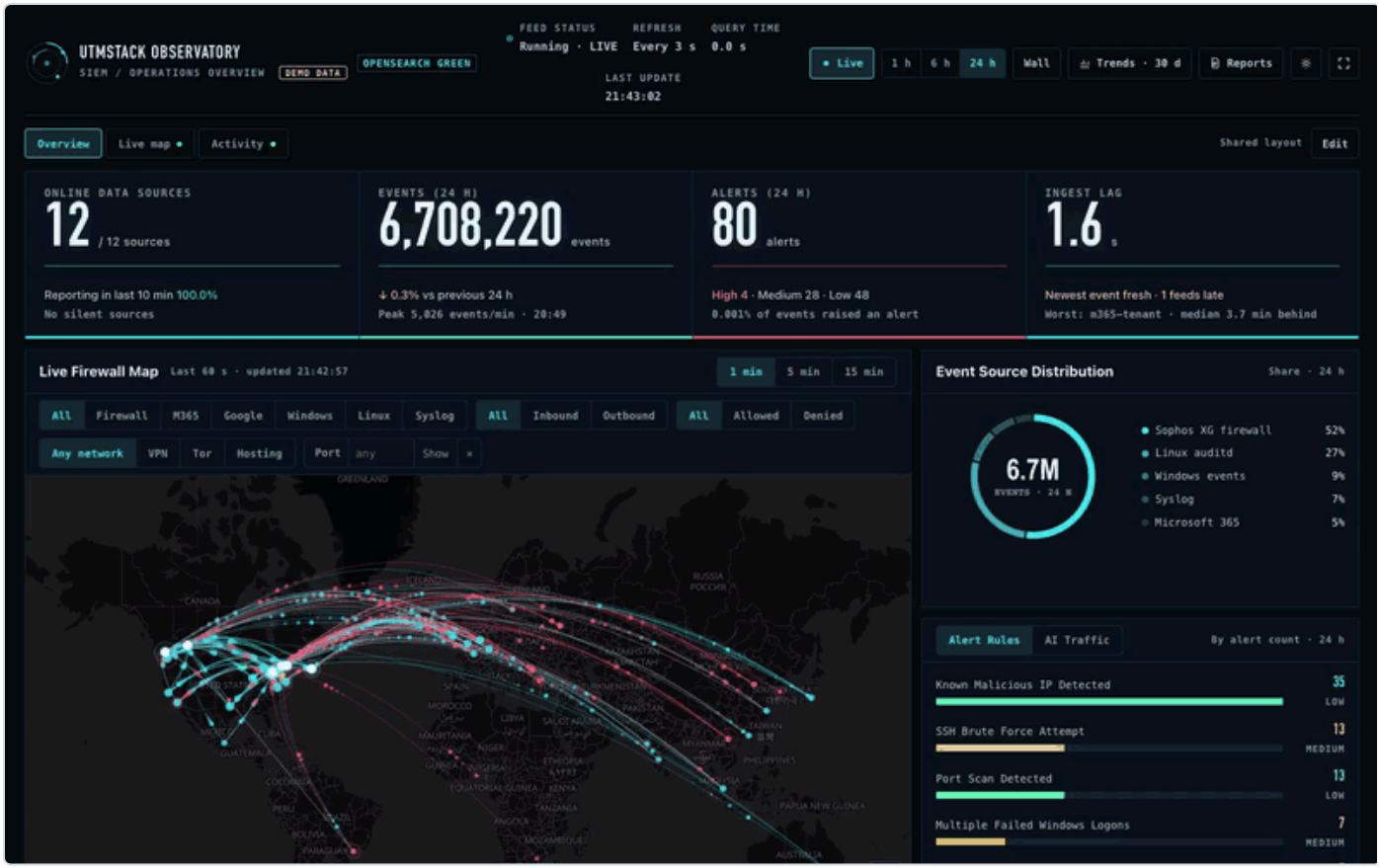


Filter by **Severity**, **Source**, **Rule** and free-text **Search**, then **Apply**. **Newer / Older** page through the results.

Click an alert to open its detail:

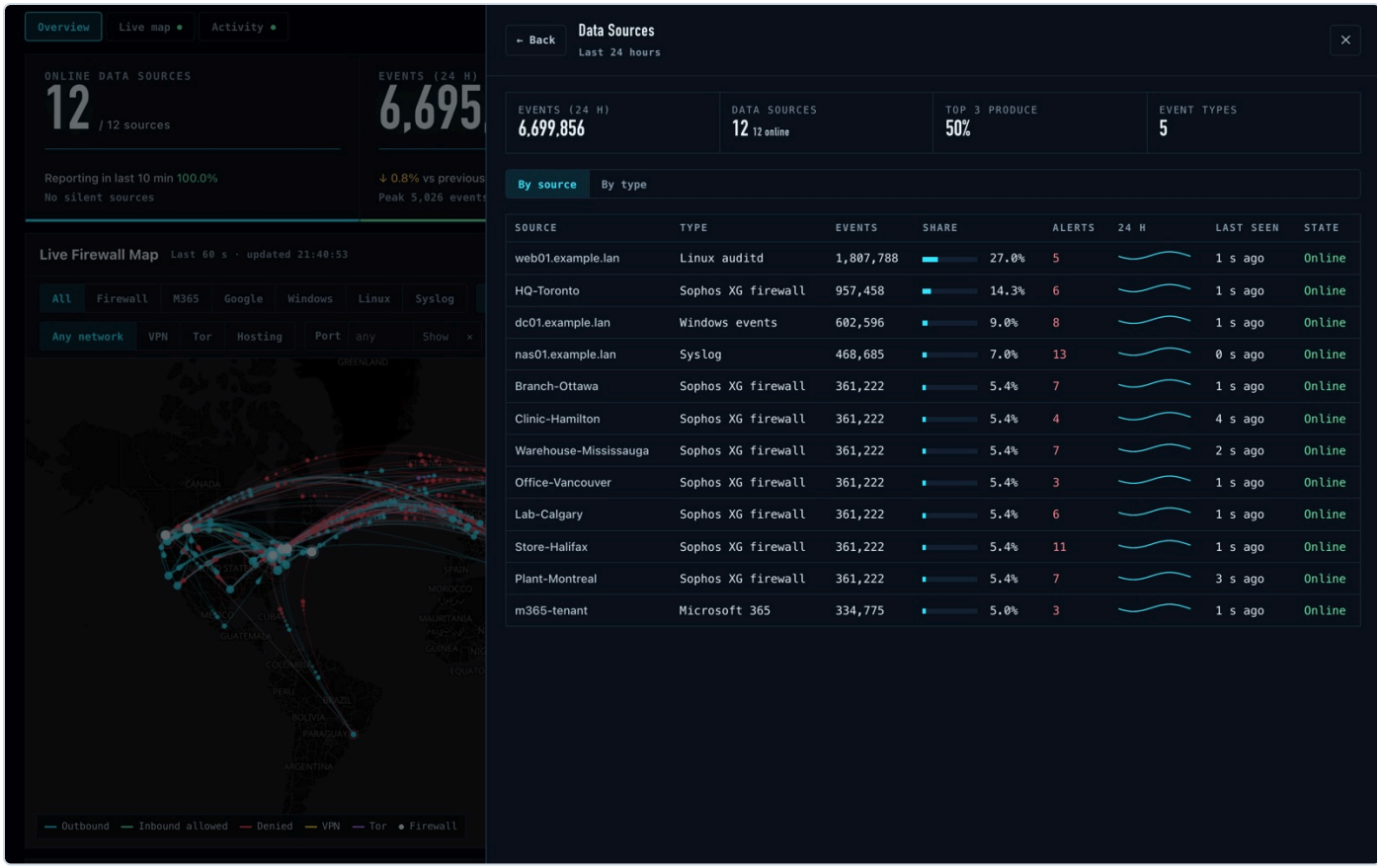


- **Recommended action**: what to do about it, in plain language.
- **Source**, **More from this source**, **Same rule**: jump to related alerts.
- **Details**: every field UTMStack stored for the alert.
- **Reputation**: live AbuseIPDB score and VirusTotal detections for the alert's public addresses, when API keys are configured.
- **WHOIS**: who owns each public address (registry, organisation, abuse contact).

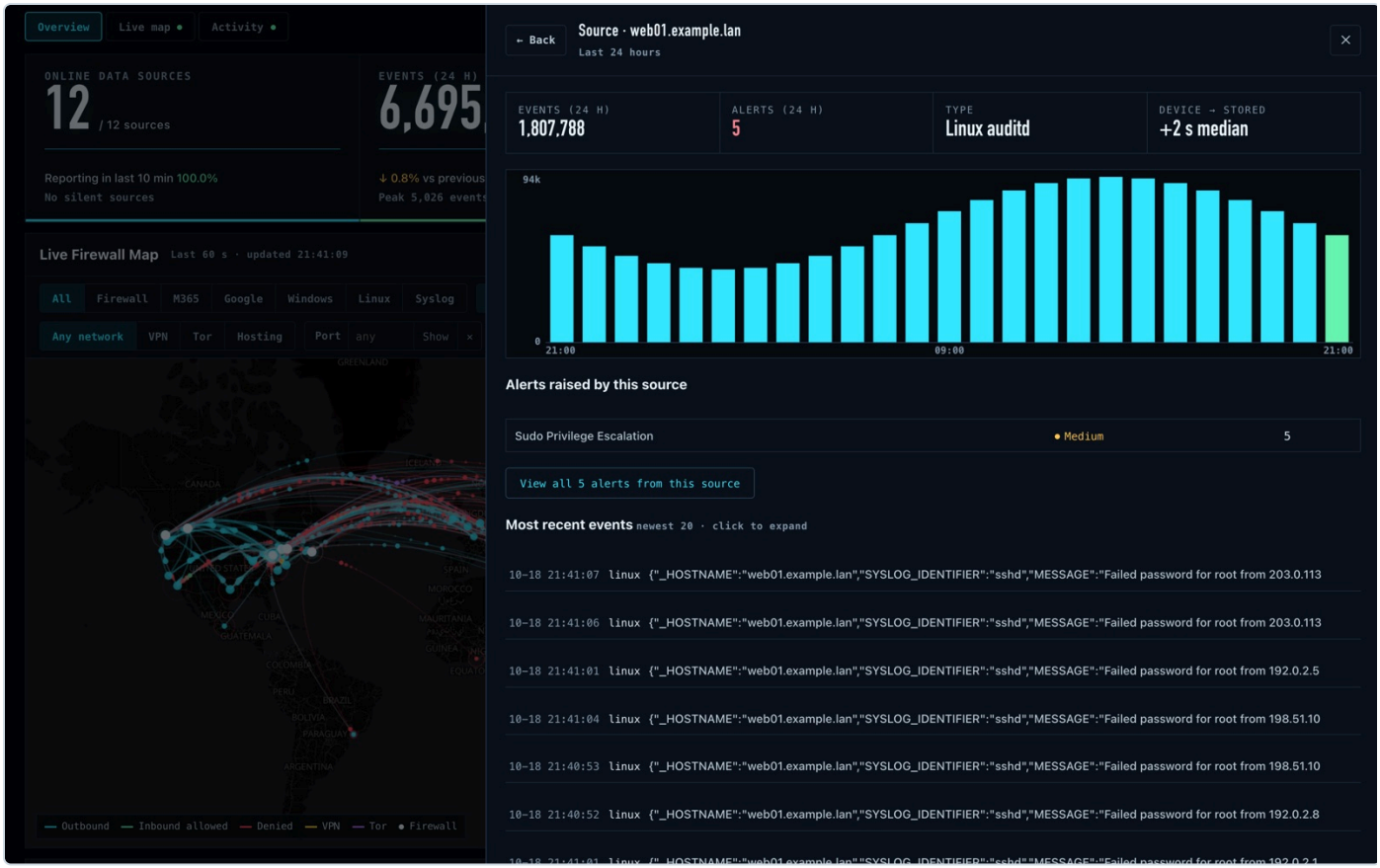


7. Data sources and ingest lag

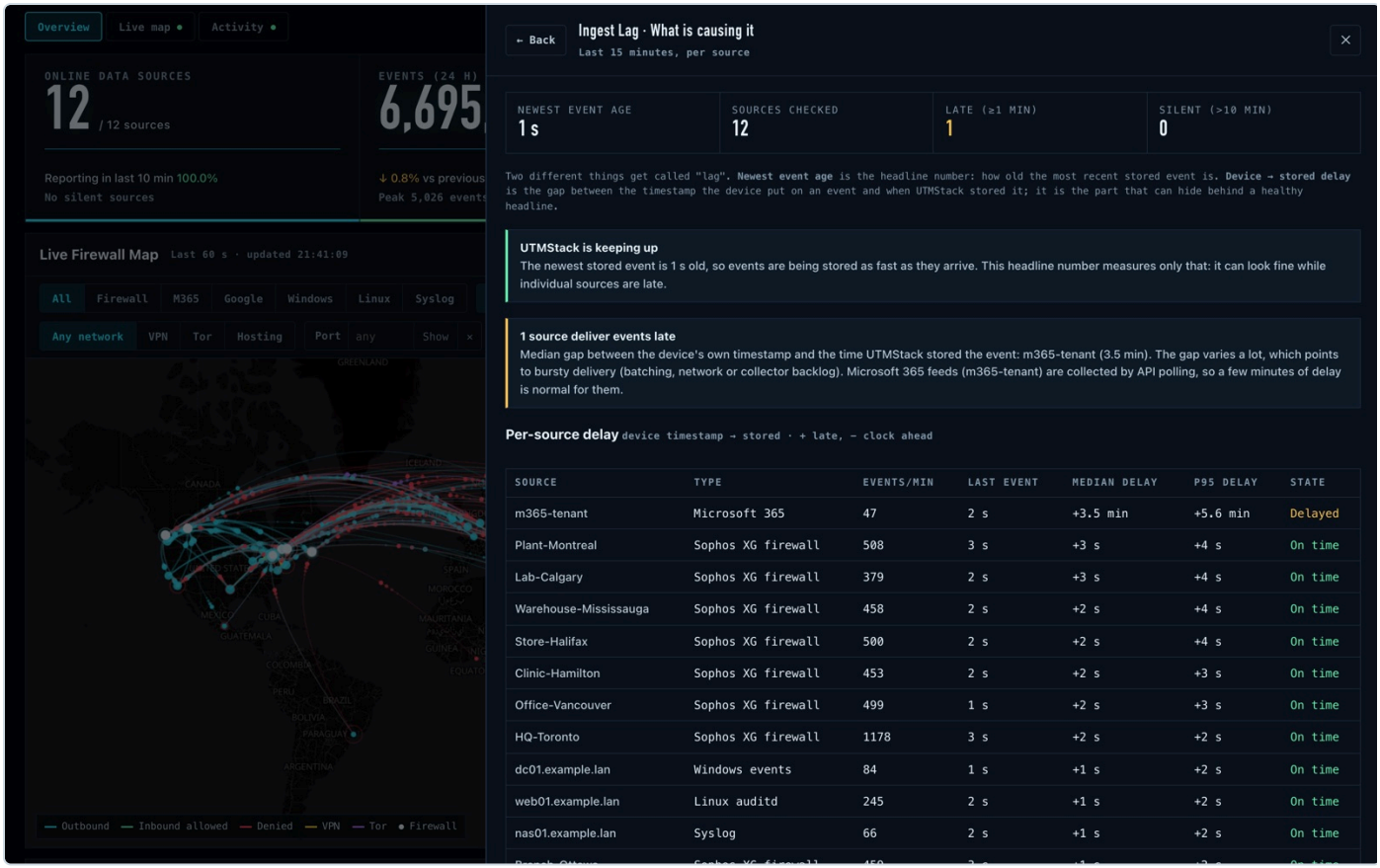
Click **Online data sources** for every source, its type, event count, share, a 24-hour sparkline, when it last reported and its status. Switch between **By source** and **By type**.



Click a source for its event history, the alerts it raised and its most recent events:

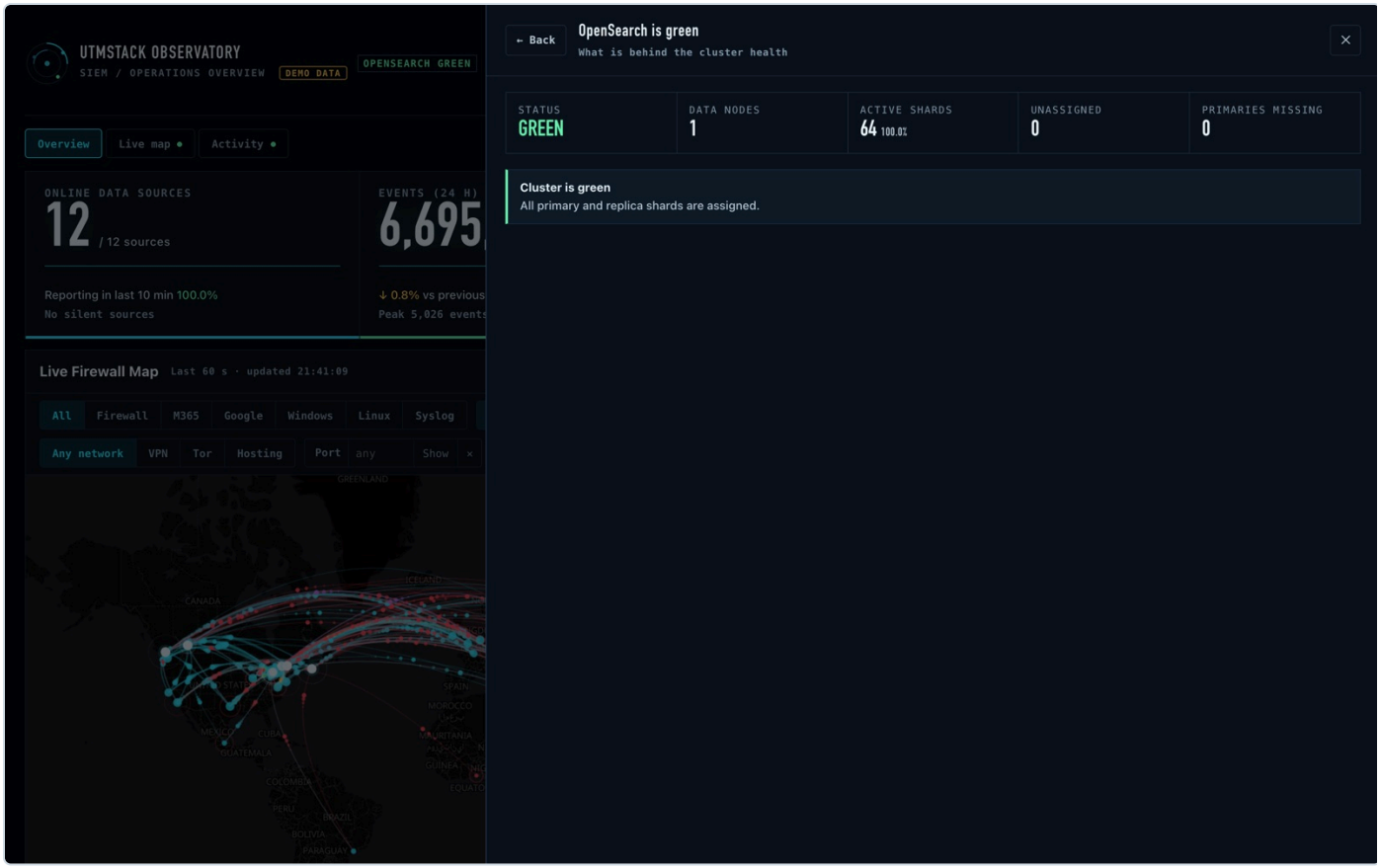


Click **Ingest lag** for the delay per source: the time between when a device logged an event and when UTMStack stored it. Firewalls often send logs in batches, and Microsoft 365 is polled, so a delay of minutes is normal for those.



8. SIEM health

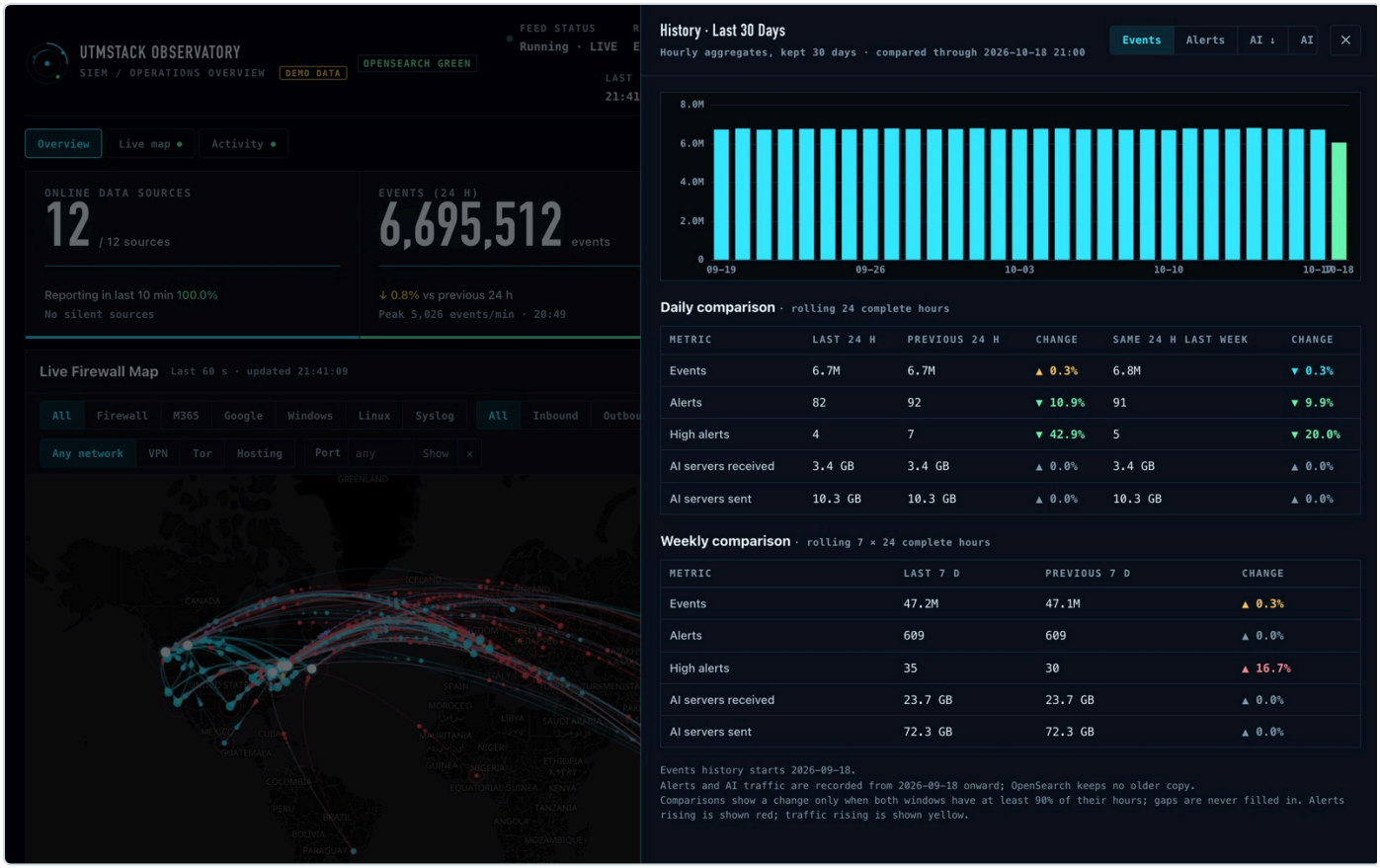
The badge next to the title shows the health of UTMStack's OpenSearch cluster. Click it for an explanation in plain language: whether data is safe, which shards are unassigned and why, and the commands that fix it. **Copy diagnostics** puts the technical details on the clipboard for your support provider.



Badge	Meaning
Green	Everything is assigned and searchable.
Yellow	All data is available, but some replica copies are missing. Common on single-node installs; usually not urgent.
Red	Some data is not searchable. Act on the advice in the panel.

9. Trends and history

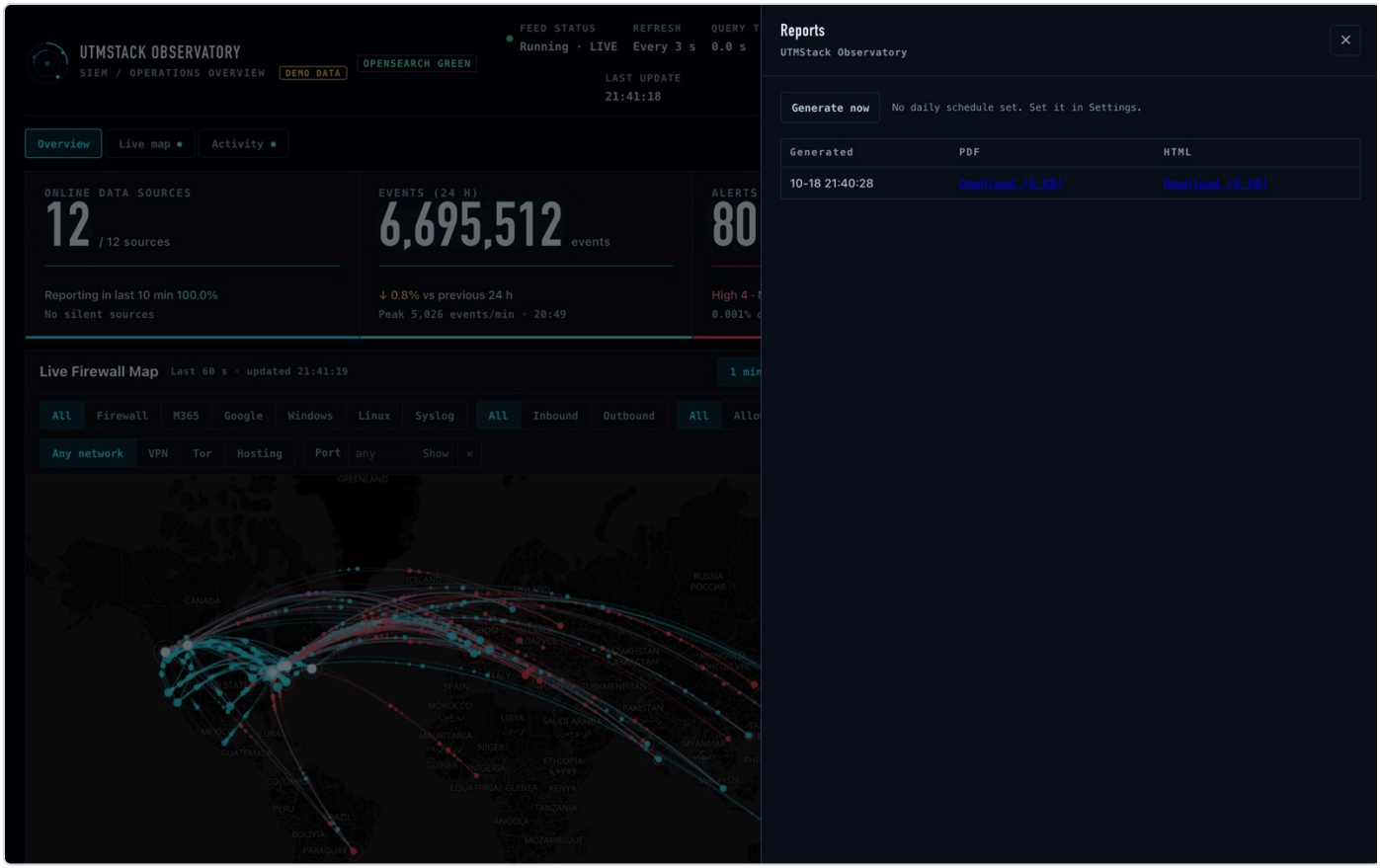
Trends · 30 d shows hourly history for the last 30 days (events, alerts and, if enabled, AI traffic), a **daily comparison** (last 24 h against the 24 h before) and a **weekly comparison** (last 7 days against the 7 before).



OpenSearch on a UTMStack server usually keeps raw logs for only about a day, so the dashboard keeps its own small history database (counts only, never raw events). History starts when the dashboard is installed. Comparisons show *n/a* or *partial* until both periods have at least 90% of their hours: about a day for daily, two weeks for weekly.

10. Reports

Reports lists generated reports, newest first, each as **PDF** and **HTML**. **Generate now** makes one for the last 24 hours: key figures and trends, top alert rules and recent alerts (with the MITRE ATT&CK technique where the rule has one), top blocked countries from the map, and storage health.



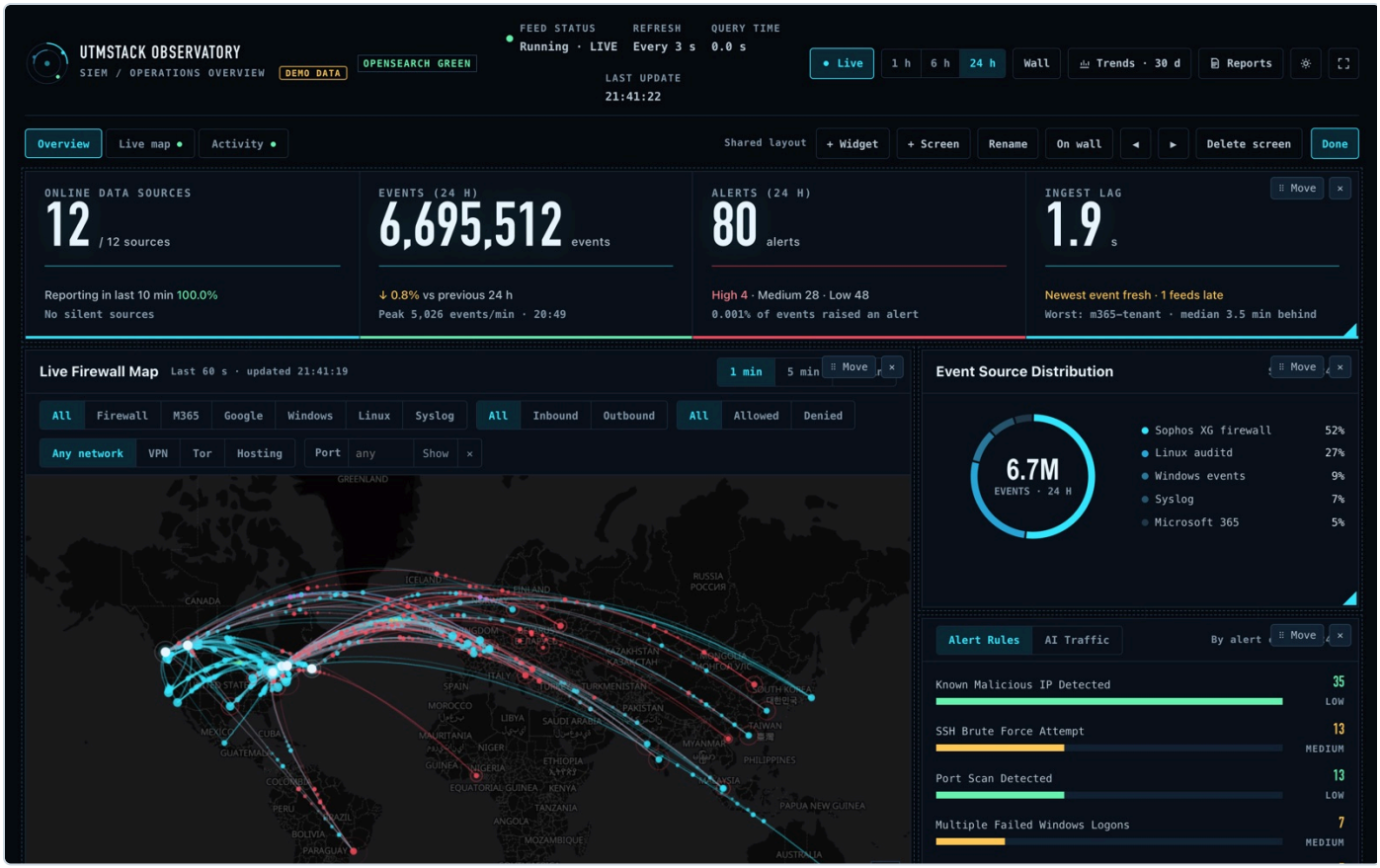
To get a report every day automatically, set **Report hour** (0–23, UTC) in the Settings panel. The last 30 reports are kept. Downloads need the dashboard login.

A sample report made from demo data is in `../media/sample-report.pdf`.

11. Screens, widgets and Wall mode

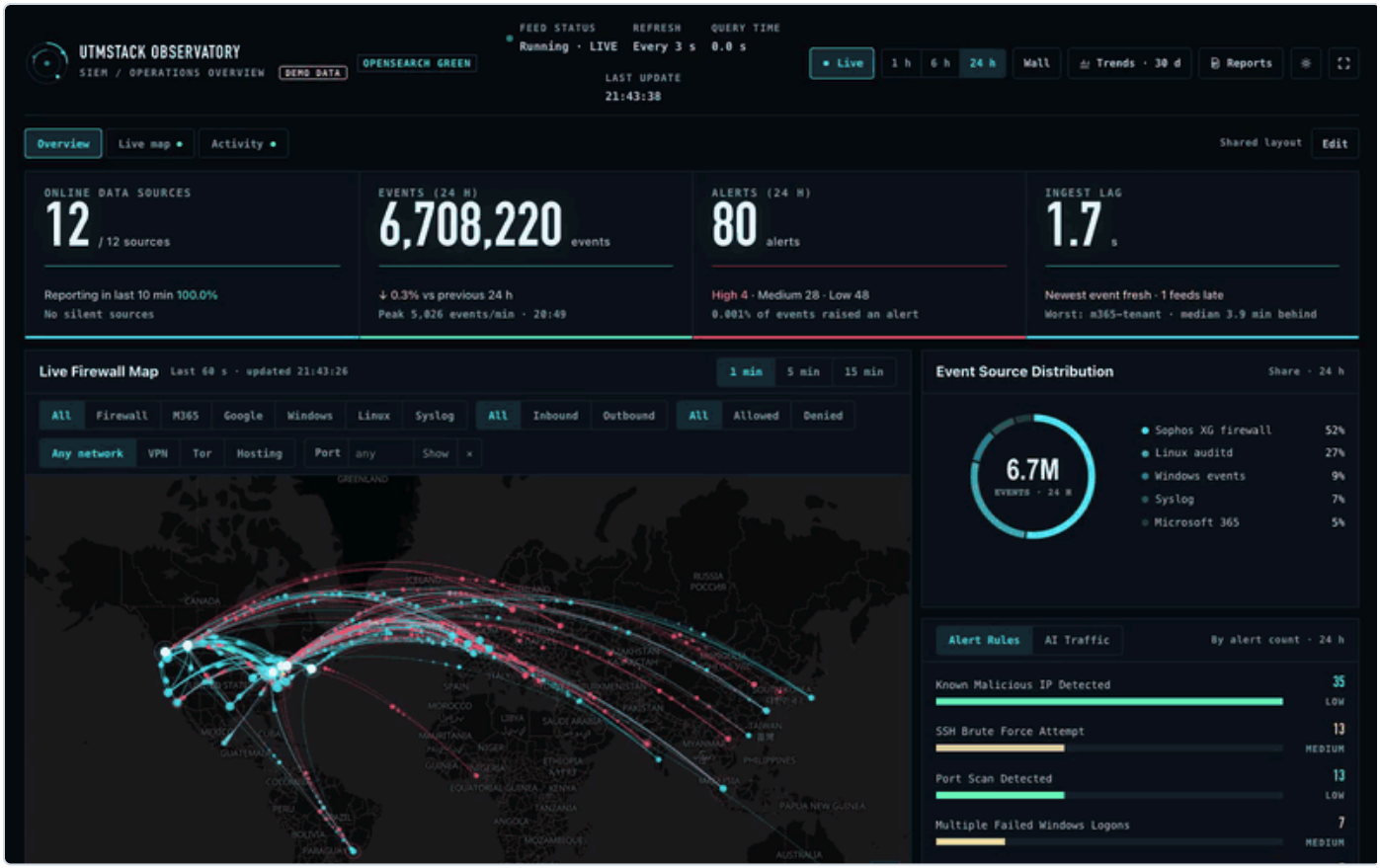
Editing the layout

Click **Edit** (right of the screen tabs).



- Drag a panel by its ⠿ **Move** handle; resize it from the corner.
- **+ Widget** opens the widget catalogue. **+ Screen** adds a new tab; **Rename** and **Delete screen** manage it.
- **On wall** includes the screen in Wall-mode rotation.
- **Save as shared default** makes your layout the one everyone sees. **Reset to shared** drops your personal changes.
- Click **Done** to leave edit mode.

Without saving as shared, layout changes are personal to your browser.



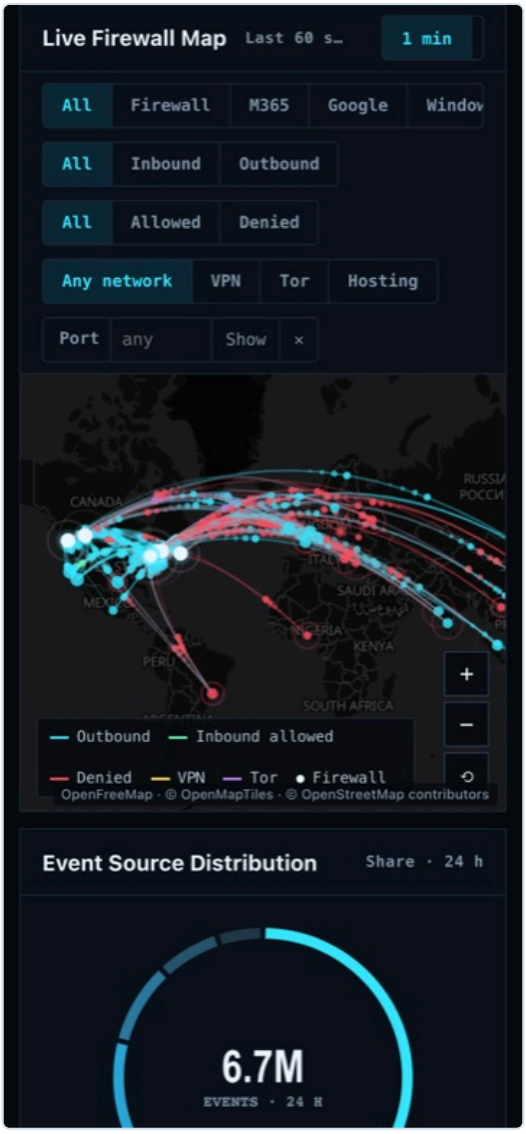
Wall mode

Wall (or open `/?wall=1`) turns the dashboard into a NOC display: full screen, rotating through the screens marked *On wall* every 30 seconds. Firewall names and WAN addresses are replaced by **Site A, Site B…** on the server before they are sent, so a wall screen in a lobby never shows them. Panels that list data-source names are left out. Press **Esc** to leave.

Phones and tablets

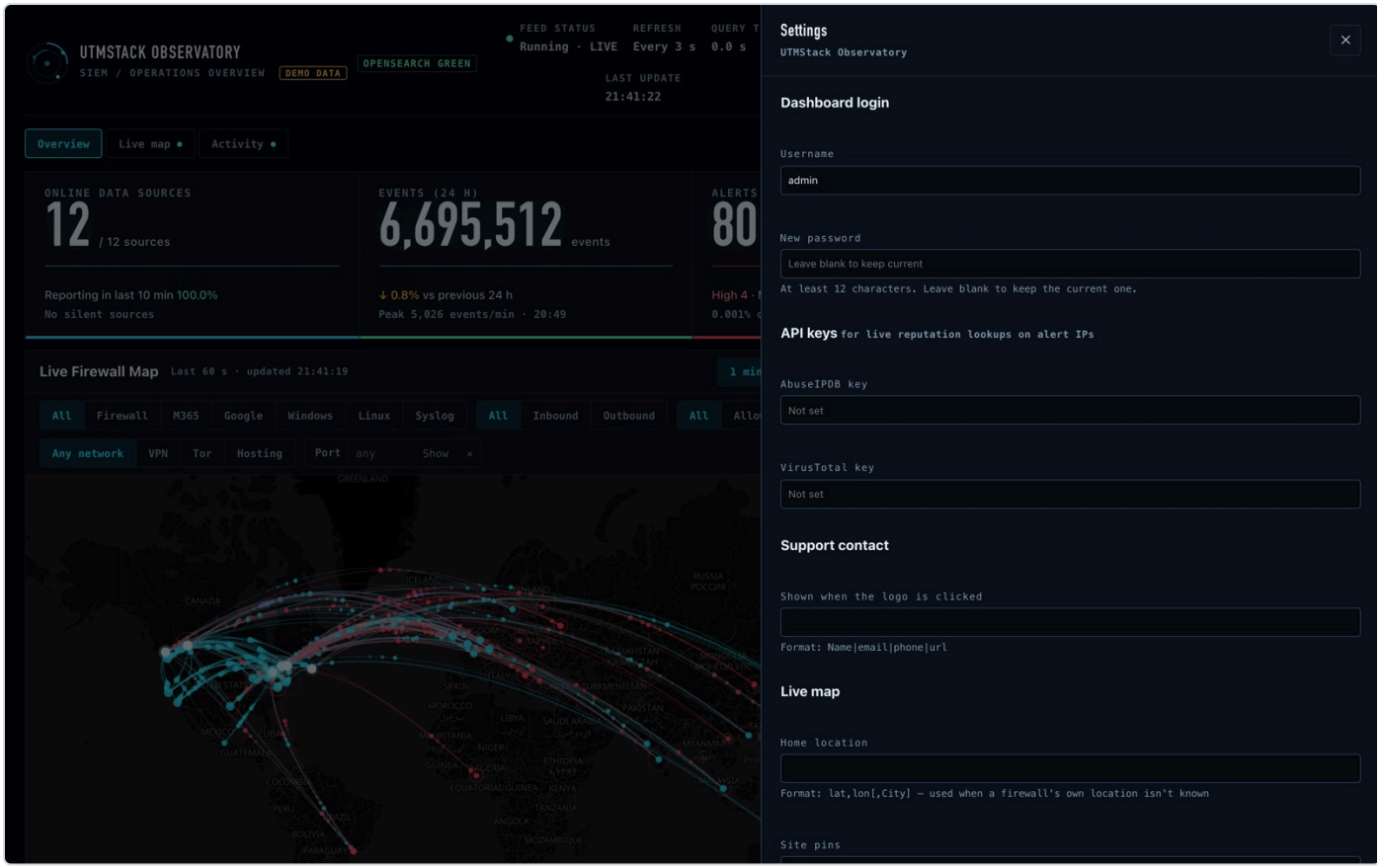
The dashboard works on a phone. Panels stack, and the map gets a taller view with touch zoom.


UTMSTACK OBSERVATORY
SIEM / OPERATIONS OVERVIEW
DEMO DATA
FEED STATUS
LAST UPDATE
Running · LIVE
21:41:30
• Live
1 h
6 h
24 h
Wall
Trends
Overview
Live map •
Activity •
Shared lay…
ONLINE DATA SOURCES
12
/ 12 sources
Reporting in last 10 min 100.0%
No silent sources
EVENTS (24 H)
6,695,512
events
↓ 0.8% vs previous 24 h
Peak 5,026 events/min · 20:49
ALERTS (24 H)
80
alerts
High 4 · Medium 28 · Low 48
0.001% of events raised an alert
INGEST LAG
2.1
s
Newest event fresh · 1 feeds late



12. Settings panel

The **gear** icon opens Settings. Changes apply immediately, with no restart. Changing the login, an API key or the support contact needs your **current password** in the *Confirm* field at the bottom. A value saved here overrides the matching installer option; clearing a field and saving falls back to the installer's value.



Section	Settings
Dashboard login	User name and password (at least 12 characters). Changing the password signs you out so you sign in again with the new one.
API keys	AbuseIPDB and VirusTotal keys for the Reputation section of alert detail. Keys are write-only: the panel shows whether a key is set, never the key itself.
Support contact	Name, email, phone and website shown when the logo is clicked.
Live map	Home location (where firewalls without a known location are drawn) and site pins (`FirewallName=lat,lon,City;203.0.113.7=lat,lon`).
AI traffic	Hosts to measure, as `name=ip,name=ip`. Empty hides the AI Traffic tab.
Reports	Daily report hour (0–23, UTC). Empty means on demand only.
Dashboard layout	Shortcut to the layout editor.

13. Demo mode

Demo mode runs the dashboard on realistic, fully synthetic data, for sales demos, training and screenshots. It needs no UTMStack and no internet connection (the street map tiles excepted).

- A fictional organisation: eight Canadian sites, Linux, Windows and storage servers, and a Microsoft 365 tenant.
- Every panel, drill-down, report, WHOIS and reputation lookup answers with sample data that moves over time like a live system.
- All public addresses come from the reserved documentation ranges `192.0.2.0/24`, `198.51.100.0/24` and `203.0.113.0/24`, so no screenshot can show a real company's address.
- A yellow **DEMO DATA** tag appears next to the title.

Start it on any machine with Python 3.10+ and `pip install -r requirements.txt`:

```
cd app
DEMO=1 DASH_PASS=choose-a-password python3 server.py
```

Then open `http://<machine>:8765`. Demo mode is off unless `DEMO=1` is set; never set it on a production install.

14. Installation and upgrades

Requirements: a UTMStack v11 server (Docker Swarm), root or sudo, and internet access for the first image build.

```
git clone https://github.com/vbeitner/utmstack-observatory.git
cd utmstack-observatory
sudo ./install.sh
```

The installer finds the UTMStack stack, reads the OpenSearch credentials UTMStack already uses, checks they work, builds the image, starts the service and prints the address and a generated password.

To **upgrade**, get the new version and run the installer again (`git pull && sudo ./install.sh`). Password, settings and history are kept.

Command	What it does
`sudo ./install.sh`	Install or upgrade
`sudo ./install.sh status`	Service state and health
`sudo ./install.sh logs`	Service logs
`sudo ./install.sh password [NEW]`	Set a password, or generate a new one. Also removes a password set in the Settings panel, so it revokes it
`sudo ./install.sh uninstall`	Remove the service, keep history and settings
`sudo ./install.sh uninstall --purge`	Remove everything, including history, settings and the image

15. Configuration reference

Installer options

Options are remembered in `/etc/utmstack-observatory/config.env`; pass one again to change it.

Option	Default	Meaning
`--port N`	`8765`	Port the dashboard listens on
`--user NAME`	`admin`	Dashboard user name
`--password PW` / `--password-file F`	generated	Dashboard password (12+ characters; the file form keeps it out of the process list)
`--tls` / `--no-tls`	on for new installs	HTTPS with a generated self-signed certificate, or plain HTTP
`--tls-cert FILE --tls-key FILE`		HTTPS with your own certificate
`--tz ZONE`	host's	Time zone for day boundaries, e.g. `America/Toronto`
`--stack NAME`	auto	UTMStack Swarm stack name
`--import-history FILE`		Seed history from an earlier install
`--sites "Name=lat,lon,City;IP=lat,lon"`		Pin firewalls the map cannot place
`--home "lat,lon,City"`	Toronto	Where unplaced firewalls are drawn
`--support "Name|email|phone|url"`	vendor	Contact shown under the logo
`--ai-hosts "name=ip,name=ip"`	off	Enable AI Traffic for these hosts
`--abuseipdb-key KEY`	off	AbuseIPDB reputation lookups
`--virustotal-key KEY`	off	VirusTotal reputation lookups
`--proxycheck-key KEY`	none	proxycheck.io key for the online location fallback
`--geoip-online-daily N`	`100`	Daily cap on online lookups; `0` disables them

Environment variables

Only needed when running `app/server.py` yourself; the installer sets them for you.

Variable	Default	Meaning
`OS_URL`		OpenSearch URL, e.g. `https://node1:9200`. Empty = use `docker exec` on the host
`OS_USER`, `OS_PASS` / `OS_PASS_FILE`		OpenSearch credentials
`DASH_USER`, `DASH_PASS` / `DASH_PASS_FILE`	`admin`	Dashboard login. Without a password it only listens on 127.0.0.1

Variable	Default	Meaning
`PORT`, `BIND`	`8765`, `0.0.0.0`	Listening port and address
`TLS_CERT`, `TLS_KEY`		Certificate and key files; enables HTTPS
`DATA_DIR`	app folder	History database, settings, layout, reports and IP data
`DISK_PATH`	`/utmstack`	Filesystem whose free space is reported
`HOST_NAME`, `TZ`		Host name shown in the resource panel; time zone
`STACK_NAME`	`utmstack`	UTMStack stack name
`AI_HOSTS`		`name=ip,name=ip`
`SITE_LOCATIONS`, `HOME_LOCATION`		Same as `--sites` and `--home`
`SUPPORT_CONTACT`		Same as `--support`
`ABUSEIPDB_KEY`, `VIRUSTOTAL_KEY` (or `_FILE`)		Reputation API keys
`PROXYCHECK_KEY`, `GEOIP_ONLINE_DAILY`	, `100`	Online location fallback
`LOGIN_MAX_FAILURES`, `LOGIN_LOCK_S`	`5`, `900`	Lockout after N failures, for S seconds
`MAX_CONNECTIONS`, `MAX_CONNECTIONS_PER_IP`	`64`, `32`	Concurrent connections in total and per address; extra ones are refused at once
`HEADER_DEADLINE_S`	`10`	A client must send its whole request header within this time
`ALLOWED_HOSTS`		Extra host names accepted when no login is configured (otherwise only localhost)
`OS_CA_FILE`, `OS_CHECK_HOSTNAME`	, `1`	Verify OpenSearch's certificate against this CA
`REQUEST_TIMEOUT_S`	`15`	Slow or idle clients are dropped after this
`DEMO`	off	`1` = demo mode, see Demo mode
`ALLOW_NO_AUTH`	off	`1` = allow listening on a network address with no password (testing only)

What is stored where

What	Where
30-day history, storage sizes	Docker volume `utmstack-observatory-data` → `/data/history.db`
Live settings, shared layout	`/data/settings.json`, `/data/layout.json`
Reports (last 30)	`/data/reports/`
IP location and VPN/Tor data	`/data/geoip/`

What	Where
Installer settings	`/etc/utmstack-observatory/config.env`
Dashboard password	`/etc/utmstack-observatory/admin.pass` (root only); a password set in Settings is stored hashed

16. Security and privacy

- **Read-only**: the dashboard only reads OpenSearch. It cannot change UTMStack.
- **Login on everything** except a minimal health check. Repeated wrong passwords lock out the source address. Changing the login, an API key or the support contact also asks for the current password, and every change is logged.
- **Container hardening**: read-only file system, all Linux capabilities dropped, unprivileged user, no Docker socket, password passed as a Docker secret.
- **Browser protections**: a Content Security Policy that allows only the dashboard's own scripts, protection against cross-site request forgery on every change, and no third-party scripts (the map library is bundled).
- **Encryption**: new installs use HTTPS with a self-signed certificate (browsers warn once; `--tls-cert` / `--tls-key` for your own). Installs from before 1.16 keep plain HTTP until re-run with `--tls`. Docker Swarm publishes the port on every interface, so restrict it with the host firewall, for example: `iptables -I DOCKER-USER -p tcp --dport 8765 ! -s 100.64.0.0/10 -j DROP`.
- **What leaves the server**: public IP addresses for WHOIS (rdap.org), reputation (AbuseIPDB, VirusTotal, only if keys are set) and the optional location fallback (proxycheck.io); periodic downloads of IP location and VPN/Tor lists; map tiles fetched by the browser from OpenFreeMap. Private addresses are never sent.
- **Stored data**: counts and aggregates only, never raw log events.

The security assessment is in `../security/SECURITY-ASSESSMENT.md`.

17. Troubleshooting

Symptom	What to check
"No traffic on the map"	Usually ingestion, not the dashboard: is the firewall still sending logs to UTMStack? Check **Data sources**.
Installer: OpenSearch rejects the credentials (HTTP 401)	UTMStack itself is broken (its backend uses the same credentials). Fix UTMStack first, then re-run the installer.
Trends shows *n/a* or *partial*	Not enough history yet: about a day for daily, two weeks for weekly comparisons.
Ingest lag ~1 s but feeds are late	The headline is the newest event; click the card for the per-source delay.
AI Traffic empty or low	It only sees traffic that crosses the firewall, not traffic inside the LAN.
Street map does not appear	The browser cannot reach `tiles.openfreemap.org`; the outline map is used instead.
Locked out	Wait 15 minutes, or restart the service, which clears the lockout.

Symptom	What to check
Anything else	`sudo ./install.sh logs` and `sudo ./install.sh status`; **Copy diagnostics** in the health panel.

18. Third-party data and licences

Component	Use	Licence
MapLibre GL JS 4.7.1	Street map rendering (bundled)	BSD-3-Clause
OpenFreeMap / OpenStreetMap	Map tiles	ODbL (© OpenStreetMap contributors)
DB-IP City Lite	IP location	CC BY 4.0, "IP Geolocation by DB-IP"
X4BNet lists_vpn	VPN and datacenter ranges	See the project's repository
Tor Project exit list	Tor exit nodes	Public data
Natural Earth (via world-atlas)	Outline map	Public domain
fpdf2	PDF reports	LGPL-3.0

Attribution for the map and IP data is shown in the dashboard itself, under the map.